QUEUING THEORY: PART 4

Fifty-fourth Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



Single-Server Models

- A **Single-Server Model** in queuing theory refers to a system where **one service facility** (or server) handles incoming customers **one at a time.** It is the most basic and widely studied queueing model.
- Represented by **Kendall's notation.** The format is **A/S/1**

  A is the arrival process, S is service time distribution and 1 means single server
- Most common is **M/M/1** queue (Poisson arrivals, exponential service, single server)
- **Application:** Bank teller or ATM, Doctor's clinic with one doctor, Call center with one line

Single server, Non-exponential Service time Distribution- Unlimited queue

- In queuing systems, service times are often assumed to follow an exponential distribution. However, in some real-life situations, service times do not follow this pattern. The **normal distribution** can be a useful alternative.
- In such queuing model, where arrivals form Poisson process and service time follow normal distribution depends on **standard deviation** for service time.

Single server, Non-exponential Service time Distribution- Unlimited queue

- The performance measures are determined as

$$p_0 = 1 - \frac{\lambda}{\mu}$$

$$L_q = \frac{\lambda^2\sigma^2 + (\lambda/\mu)^2}{2(1-\lambda/\mu)} \qquad L_s = L_q + \frac{\lambda}{\mu}$$

$$W_q = \frac{L_q}{\lambda} \qquad W_s = W_q + \frac{1}{\mu}$$

Single server, Non-exponential Service time Distribution- Unlimited queue

Example: At a shop, customers arrive at a mean rate of 5 per hour. Customers are served at a mean rate of 6 per hour. The service time has a standard deviation of 0.06 hour. Determine

(a) expected number of customers in the queue

(b) expected number of customers in the system

(c) expected waiting time in queue

(d) expected waiting time in system

Single server, Non-exponential Service time Distribution- Unlimited queue

Given:

$\lambda = 5$, $\mu = 6$, $\sigma = 0.06$

(a) expected number of customers in the queue

$$L_q = \frac{\lambda^2\sigma^2 + (\lambda/\mu)^2}{2(1-\lambda/\mu)} = \frac{5^2 \times 0.06^2 + (5/6)^2}{2(1-5/6)} = 2.35$$

(b) expected number of customers in the system

$$L_s = L_q + \frac{\lambda}{\mu} = 2.35 + \frac{5}{6} = 3.18$$

Single server, Non-exponential Service time Distribution- Unlimited queue

Given:

λ = 5, μ = 6, σ = 0.06

(c) expected waiting time in queue

$$W_q = \frac{L_q}{\lambda} = \frac{2.35}{5} = 0.47$$

(d) expected waiting time in system

$$W_s = W_q + \frac{1}{\mu} = 0.47 + \frac{1}{6} = 0.64$$

Single server, Constant Service time - Unlimited queue

- **Queue Type**: Single-server queue with unlimited waiting capacity.
- **Arrival Pattern:** Typically modeled using a Poisson process (random arrivals).
- **Service Time:** Constant (deterministic), meaning every customer is served in exactly the same amount of time.

Notation: Represented as M/D/1 in Kendall's notation

M = Markovian (Poisson) arrivals, D = Deterministic (constant) service time, 1 = One server

- **Applications**: Used where service processes are highly automated or mechanized (e.g., ticket vending machines, automated toll booths).

Single server, Constant Service time - Unlimited queue

- For constant service time, standard deviation (σ) is 0.
- Substituting σ=0, in

$$L_q = \frac{\lambda^2\sigma^2 + (\lambda/\mu)^2}{2(1-\lambda/\mu)}$$

Length of queue when service time is not constant

the length of queue, L_q is obtained as

$$L_q = \frac{(\lambda/\mu)^2}{2(1-\lambda/\mu)}$$

$$L_s = L_q + \frac{\lambda}{\mu}$$

$$W_q = \frac{L_q}{\lambda}$$

$$W_s = W_q + \frac{1}{\mu}$$

Single server, Constant Service time - Unlimited queue

- **Example:** At a recycling facility, truck drivers delivering plastic materials currently experience an average waiting time of 10 minutes before unloading. The cost of idle time (time wasted in queue) for both driver and truck is Rs 300 per hour. Management has decided to install a new unloading device that would process truck loads at a constant rate of 10 trucks per hour, with an operational cost of Rs 15 per truck unloaded. Truck arrivals follow a Poisson distribution with an average arrival rate of 8 trucks per hour. Is the new unloading system profitable?

Single server, Constant Service time - Unlimited queue

Solution:

- Waiting cost = Waiting time × Cost of waiting

  = (10/60) × 300 = Rs 50 per trip

In new unloading system,

$\mu = 10, \quad \lambda = 8$

$$W_q = \frac{L_q}{\lambda} = \frac{1}{\lambda} \times \frac{(\lambda/\mu)^2}{2(1-\lambda/\mu)}$$

$$= \frac{1}{8} \times \frac{(8/10)^2}{2(1-8/10)} = \frac{1}{5}\text{hour} = 12\,\text{mins}$$

Single server, Constant Service time - Unlimited queue

Solution:

- Waiting cost per trip in new unloading system= (1/5) × 300 = Rs 60 per trip
- Loss = Rs (60–50) = Rs 10/- per trip
- Hence, the management should not opt for new unloading system.

Service level Cost Model

Service level is a function of two conflicting costs:

(i) Cost of offering the service to the customers

(ii) Cost of delay in offering service to the customers

- Hence, an optimum service level that must be maintained by a queuing system can be determined by determining the optimum level of service rate (μ).
- This rate should offer service and number of service facilities (s, servers) in a way that avoids excessive delay in offering the service.

Optimum Service rate

Let

C_1 = cost per server per unit of time, i.e. cost per unit of time having a server available

C_2 = cost of waiting per customer per unit time

T_C = total expected cost of waiting and service per time

Then, the cost equation can be written as

T_C = (Service rate × Cost per service per unit time) + Number of customers in the system

x Waiting cost per customer per unit time)

$$= \mu C_1 + L_s C_2$$

$$= \mu C_1 + \frac{\lambda}{\mu - \lambda} C_2$$

For single server queuing model considering exponential service and unlimited queue

$$L_s = \frac{\lambda}{\mu - \lambda}$$

Optimum Service rate

As service rate is measured in continuous units of time, the optimum value can be determined by taking the derivative of total cost with respect to μ and equating it to zero to identify the point of minimum cost

$$\frac{\mathrm{d}T_c}{\mathrm{d}\mu} = 0$$

$$\text{or } \frac{\mathrm{d}}{\mathrm{d}\mu}\left(\mu C_1 + \frac{\lambda}{\mu - \lambda} C_2\right) = 0$$

$$\text{or } C_1 - \frac{\lambda C_2}{(\mu - \lambda)^2} = 0$$

$$\text{or } \mu = \lambda + \sqrt{\frac{C_2}{C_1}\lambda}$$

Optimum Service rate

- Customer arrives at a service store on an average rate of 32 customers per hour. Currently, a single cashier serves customers at a rate of 40 customers per hour. The store owner estimates that every additional minute a customer spends in the system (including waiting and service time) results in a cost of Rs. 5. An assistant can be employed to support the cashier (at the rate of Rs. 5/hour), thereby increasing the total service rate to 120 customers per hour. Determine the optimal service rate (in customers per hour) that would minimize the total customer waiting cost. Justify the requirement of assistant.

Optimum Service rate

- Given, λ = 32 customers per hour, μ = 40 customers per hour, C_1 = Rs. 5/-

Cost per additional customer service = Rs 5

Waiting cost (C_2) = Expected number of customers in the system x Hourly waiting cost

$$= \frac{\lambda}{\mu - \lambda} \times 5 \times 60$$

$$= \frac{32}{40 - 32} \times 5 \times 60 = \text{Rs. } 1200 / \text{hr}$$

New value of service rate

$$\mu = \lambda + \sqrt{\frac{C_2}{C_1}\lambda} = 32 + \sqrt{\frac{1200}{5} \times 32} = 119 \text{ customers per hour}$$

Hence, the requirement of assistant is justified.

Service levels can be discrete

- A company operates a customer service center where:
- Customers arrive at a rate of **λ = 10 customers per hour**.
- The cost of customer waiting time is **$5 per customer per hour**.
- The cost of providing service is **$20 per server per hour**.
- The service rate **μ** (customers served per hour per server) can be increased by hiring more or faster servers, but this increases the cost.
- The objective is to **minimize the total cost**:
- **Total Cost (TC) = Service Cost + Waiting Cost**

WE can try various service levels, for example 5 customers/hour, 10 customers per hour etc.